STACKS, DIFFERENTIAL GRADED SHEAVES: LEMMA-FIRST-PROPERTY-DG-INJECTIVE

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 3994-4010 of sources/stacks-sdga/pinned-original-source.tex; proof span is lines 4012-4021, the complete author proof (8 body lines).

Editorial note. Navigation stubs: 0 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

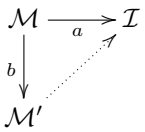
Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 46 of queue_sdga.json (pb=452, stmt 3994-4010, proof 4012-4021).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Lemma 0.1. *Let $(\mathcal{C}, \mathcal{O})$ be a ringed site. Let $(\mathcal{A}, d)$ be a sheaf of differential graded algebras on $(\mathcal{C}, \mathcal{O})$. Let $\mathcal{I}$ be a K-injective and graded injective object of Mod$(\mathcal{A}, d)$. For every solid diagram in Mod$(\mathcal{A}, d)$*



where b is injective and $\mathcal{M}$ is acyclic a dotted arrow exists making the diagram commute.

Proof. Since $\mathcal{M}$ is acyclic and $\mathcal{I}$ is K-injective, there exists a graded $\mathcal{A}$-module map $h : \mathcal{M} \to \mathcal{I}$ of degree -1 such that $a = \mathrm{d}(h)$. Since $\mathcal{I}$ is graded injective and b is injective, there exists a graded $\mathcal{A}$-module map $h' : \mathcal{M}' \to \mathcal{I}$ of degree -1 such that $h = h' \circ b$. Then we can take $a' = \mathrm{d}(h')$ as the dotted arrow. □

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